

Which CHIP Is The FASTEST



Dilin Anand is executive editor at EFN. He is B.Tech in ECE from University of Calicut, and MBA from Christ University, Bengaluru

Intel launched its eighth-generation processors in August this year, but these weren't the 10nm process chips most techies were looking for. The chip-maker had originally forecasted that it would have 10nm chips ready at the end of 2016, which is roughly two years after the launch of their 14nm process based chips as per the usual tick-tock cycle. But we all know hardware is hard, especially when it needs to be engineered and fabricated at the nanometre scale!

For its part, Intel says that it's looking at things from a user's perspective instead of a strictly technical one, and that 40 per cent upgrade in performance is significant enough for a generational rebranding. But is it enough to beat the other high-performance contenders like AMD's Ryzen? Or more importantly, which one would be an optimal pick for your next high-performance computing project?

Let's take a look.

Do you need peak graphics performance for virtual reality?

If yes, you should be looking at central processing units (CPUs) that can support two-way scalable link interface (SLI) and up to 64GB of DRAM. SLI is a multiple graphics processing unit (GPU) technology that allows users to increase graphics computing capability on a computer. How does it work? Parallel

processing is used to link two or more video cards together in order to produce a single much more powerful output.

Currently, Ryzen 7 1800X chip tops AMD's Ryzen 7 range. It has eight processor cores, 4GHz Turbo Core clock speed and Socket AM4 package. The AM4 is a microprocessor socket from AMD that has 1331 pin slots and supports double-data-rate DDR4 memory.

While this chip does have a thermal design power (TDP) of 95W, a dual-GPU setup on an AM4 motherboard platform will completely take care of things as long as you keep the processor under 95°C. The chip uses a new cooling system. In fact there are very good cooling options available from brands like AMD's Wraith, Thermalright and Swiftech. AGESA (AMD Generic Encapsulated Software Architecture) software updates will also continue to improve performance as the newer versions keep coming in.

An alternative is to use Kaby Lake-X based Core i7-7740X high-end chip, which when used with X299 motherboards can deliver some serious computing power. Launched earlier this year, this chip has four cores and eight threads, and 4.5GHz of maximum Turbo clock speed. However, its TDP is 112W. The maximum temperature allowed on the processor die is 100°C, so your cooling system better be a really good one.